LAB TASKS

TASK # 01:

Write a C++ program that accepts an amount from the user and calculates the minimum number of notes required to make that amount using the available denominations (Rs. 5000, 1000, 500, 100, 20, 10, 5, 2, and 1). The program should determine and display the fewest notes needed for the given amount, based on the available denominations.

Input:

A positive integer representing the amount (in Rs.) for which the minimum number of notes needs to be calculated.

Output:

Display the minimum number of notes required, along with the denomination of each note used.

Example:

Input:

Amount = 5786

Output:

Minimum notes required for Rs. 5786 are:

5000 x 1

500 x 1

100 x 2

20 x 4

5 x 1

1 x 1

TASK # 02

Write a C++ program to input a number from user and print a diamond pattern of alphabet characters according to input.

Input and Output:

```
Enter the Number of Rows: 5
    A
   ABC
  ABCDE
 ABCDEFG
ABCDEFGHI
 ABCDEFG
  ABCDE
   ABC
    A
```

TASK # 03

Write a C++ program that defines a function named findPrimesInRange, which takes two numbers as input and calculates all the prime numbers between them (exclusive). The program should print all the prime numbers found within the given range.

Input:

Two integers, start and end, where the program will calculate and display all prime numbers between start and end (excluding start and end).

Output:

A list of prime numbers between the given start and end.

Example

Input:

Enter the start number: 10
Enter the end number: 50

Output

Prime numbers between 10 and 50 are:
11 13 17 19 23 29 31 37 41 43 47

TASK # 04
Write a C++ program that performs the following tasks:

1. Asks the user to input size and values for two arrays.
2. Swaps the values of both arrays.
3. Finds the largest element from the first array (after swapping).
4. Finds the second largest element from the second array (after swapping).
5. Finds the unique element (element that appears only once) in the first array (after swapping) and if all values in the array are unique, return the value at index 0.

Input:

- Two arrays of integers (1D) provided by the user. The program ensures that both arrays have the same size.

Output:

- The new swapped arrays.
- The largest element from the first swapped array.
- The second largest element from the second swapped array.
- The unique element in the first swapped array.

Example:

Input:

Enter size of arrays: 5
Enter 5 elements for the first array: 2 7 3 5 9
Enter 5 elements for the second array: 4 2 6 7 8

Output:

Arrays after swapping:
First array: 4 2 6 7 8
Second array: 2 7 3 5 9

Largest element in first array (after swapping): 8
Second largest element in second array (after swapping): 7
Unique element in first array (after swapping): 4

TASK # 05
Write a C++ program to multiply two matrices using a **function** named multiplyMatrices. The program should take two 2-dimensional arrays (matrices) as input, validate the multiplication rule, and calculate the product of the matrices. Use the following instructions:

Rules for Matrix Multiplication:

1. The **number of columns** in the first matrix must equal the **number of rows** in the second matrix.

Input:

1. Dimensions of the first matrix **A** (m rows and n columns).
2. Dimensions of the second matrix **B** (n rows and p columns).
3. Elements of both matrices **A** and **B**.

Output:

1. The resulting matrix **C** after multiplication.

Example:

Input:

Enter rows and columns of the first matrix: 2 3
Enter rows and columns of the second matrix: 3 2

Enter elements of the first matrix:
1 2 3
4 5 6

Enter elements of the second matrix:
7 8
9 10
11 12

Output:
Resultant Matrix after Multiplication:
58 64
139 154